Maritime & Coastguard Agency

Guidance

Annex A

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Contents

Annex A - Hazards, risks and ways of avoiding them

The table lists the agents, processes and work taken from the annex to the European directive on the protection of young people at work (94/33/EC) which the European Union considers are likely to give rise to dangers to young people (see paragraph 4.4 of this Notice).

Where these are considered likely to apply to young workers on ships, advice is given on what risks may arise, and how they should be avoided.

Work objectively beyond physical or psychological capacity

Lists of agents, processes and work	Risk	How to avoid risk
Physical capacity: e.g. the lifting, moving or carrying of heavy loads or objects or similar work which is beyond their physical capacity	Accidents, injuries and or musculoskeletal disorders which can occur in jobs that require repetitive or forceful movements, particularly in association with awkward posture or insufficient recovery time	The risk assessment should take account of the physique and general health, age and experience. Training and appropriate supervision should be provided.
Work the pace of which is determined by machinery and which involves payment by results	Young people may be more at risk as their muscle stretch may not be fully developed, and they may be less skilled in handling techniques or in pacing the work according to capacity. They may also be more subject to peer pressure to take on tasks that are too much for them or to work more quickly.	The risk assessments should take account of age and experience. Training and appropriate supervision should be provided.
Psychological capacity	Although there will large individual differences in the psychological capacity	The risk assessment should focus on critical tasks which rely on skill,

Lists of agents, processes and work	Risk	How to avoid risk
	of young people based on differences in training, experience, skills, personality and attitudes in the vast majority of jobs there is no difference in the kind of mental and social skills used by young people and adults.	experience and an understanding of the task requirements. Training and effective supervision should be provided, particularly where the young person might be using machinery with exposed dangerous parts.

Work involving harmful exposure to agents which are toxic, carcinogenic, cause heritable genetic damage, or harm to the unborn child, or which in any other way chronically affect human health

Lists of agents, processes and work	Risk	How to avoid risk
Physical agents: Work in high pressure atmospheres	Not relevant on fishing vessels	
Noise	There is no evidence that young people face greater risk of damaged hearing from exposure to noise than other workers.	Reference documents: Merchant Shipping and Fishing Vessels (Control of Noise at Work) Regulations 2007 and MGN 658 (M+F). Compliance with the regulations will protect the

Lists of agents, processes and work	Risk	How to avoid risk
		hearing of most young people. Ensure that a competent person supervises the wearing of ear protection to ensure it is worn properly, during exposure to loud noise.
Hand-arm vibration	There is no evidence that young people face greater risk of developing hand-arm vibration syndrome (vibration white finger) following exposure to hand-arm vibration than other workers. However there is an increased risk in the onset of non-occupational Raynaud's disease during adolescence which can give similar symptoms to vibration white-finger. Young persons with non-occupational Raynaud's disease should not be exposed to hand-arm vibration.	Reference documents: the Merchant Shipping and Fishing Vessels (Control of Vibration at Work) Regulations 2007 and MGN 353 (M+F) Amendment 1. Action should be taken to protect young persons where exposure exceeds an acceleration of 2.5 m/s2. Employers will need to consider a programme to control the significant risks identified in the risk assessment including identification of hazardous equipment/tasks; limiting exposure to 1 m/s2 normalised over 8 hours (A(8)), by reducing either time of exposure and/or vibration level; providing competent supervision; and health surveillance.
Whole body vibration	Regular exposure to low frequency vibration, or to shocks, for example working in fast rescue craft, may be associated with back pain and other spinal disorders.	Reference documents: Merchant Shipping and Fishing Vessels (Control of Vibration at Work) Regulations 2007; MGN 353 (M+F) Amendment 1

Lists of agents, processes and work	Risk	How to avoid risk
	Younger workers may be at increased risk of damage to the spine as the strength of the muscles is still developing and the bones do not fully mature until around the age of 25.	and MGN 436 (M+F) Amendment 2. Action should be taken to protect young persons when exposure exceeds an acceleration of 0.5 m/s2. Employers will need to consider a programme to control the significant risks identified in the risk assessment including: identification of hazardous equipment/tasks; limiting exposure by reducing the time and/or level; producing information and training on how to minimise the risk; and health monitoring.
Biological agents (micro-organisms)	Despite physical and psychological immaturity, young persons are no more likely to contract infections from biological agents than adults. Like any other workers they may be at greater risk if they suffer from any other disease, are taking medication or are pregnant.	Reference documents: the Merchant Shipping and Fishing Vessels (Health and Safety at Work) (Biological Agents) Regulations 2010 and MSN 1889(M+F) Amendment 2. Precautions should be taken to protect all workers against risk of infection at work and of acquiring an allergy to certain microbes, regardless of their age or state of health. Following a risk assessment approach, control measures can be as

Lists of agents, processes and work	Risk	How to avoid risk
		simple as maintaining high standards of hygiene i.e. hand washing or use of gloves. Where available, vaccination should be offered as a supplement to procedural or physical controls.
Chemical agents: Very toxic, toxic, harmful, corrosive and irritant substances	Young people are not physiologically at any greater risk from exposure to such substances than anyone else. However, young people may lack awareness of hazards and risks to their health.	Reference documents: Merchant Shipping and Fishing Vessels (Health and Safety at Work) (Chemical Agents) Regulations 2010 as amended and Merchant Shipping Notice MSN 1888 (M+F) Amendment 2. Employers should assess the health risks to young people, arising from work with such substances, and where appropriate use a safer substance or change the process to avoid its use. Where information is supplied with the substance, for example under regulations on classification, labelling and packaging of hazardous substances (CLP) or the Control of Substances Hazardous to Health (COSHH) Regulations, particular attention should be paid to such information.

Lists of agents, processes and work	Risk	How to avoid risk
		Workers should be given any relevant information affecting their health and safety, instruction and training in the safe handling and use of the substance and be provided with adequate supervision within a safe system of work.
Carcinogenic substances	Some substances (carcinogens) may cause cancer. They need special consideration because of that property – they have no special effect on young people.	Reference documents: Merchant Shipping and Fishing Vessels (Health and Safety at Work) (Carcinogens and Mutagens) Regulations 2007 as amended and Marine Guidance Note MGN 624 (M+F). Many of these substances can be identified from the label or safety data sheet (where supplied) for the substance, which will say “May cause cancer”. Exposure is to be reduced to as low a level as reasonably practicable.

Lists of agents, processes and work	Risk	How to avoid risk
Mutagens	Some substances may impair people's ability to have children or may damage the unborn child.	Reference document: MSN 1890 (M+F) Amendment 2 Exposure is to be reduced to as low a level as reasonably practicable.
Substances causing allergic reactions	Some substances can cause allergic reactions in people. This may give them dermatitis or asthma. These substances do not affect young people any differently from adults.	HSE guidance on preventing asthma at work and dermatitis, gives practical advice on preventing risk to all workers.
Lead and lead compounds	Young people are not physiologically at any greater risk from exposure to lead and its compounds than anyone else. Lead and its inorganic compounds are known to produce diverse biological effects in humans depending on the exposure level. These range from minor biochemical changes in the blood, to severe irreversible or life threatening disruption of body processes, in particular the nervous system and the kidneys. There are also concerns about the effects of lead on the quality of semen and on the unborn child.	See above on chemical agents. Lead may be found in some paints. Guidance on precautions to be taken while painting or cleaning or rubbing down painted surfaces, are contained in Chapter 25 of the Code of Safe Working Practices for Merchant Seafarers. Employers should ensure that they adequately control the exposure of young people to lead and its compounds. Special attention should be paid to the provision of information, instruction and training, and to the

Lists of agents, processes and work	Risk	How to avoid risk
	The toxic effects of lead alkyls are primarily neurological or psychiatric. Symptoms include agitation, insomnia, dizziness, tremors and delirium, which can progress to mania, coma and death. These symptoms are accompanied by nausea, vomiting and abdominal pain. The actual risk can only be determined following a risk assessment of the particular circumstances under which there is exposure at the place of work. However, young people may not appreciate the dangers to their health or they may not understand or follow instructions properly because of their immaturity	provision of adequate supervision within a safe system of work.
Asbestos	Young people are not physiologically at any greater risk from exposure to asbestos than anyone else. However, young people may not be aware of the hazards and risk to their health or follow instructions properly because of their immaturity. Exposure to asbestos fibres causes three serious diseases; - Mesothelioma (a cancer of the lung lining) - Lung cancer (indistinguishable from cancers caused by other	Reference documents: Merchant Shipping and Fishing Vessels (Health and Safety at Work) (Asbestos) Regulations 2010 as amended and MGN 429(M+F) and Merchant Shipping and Fishing Vessels (Health and Safety at Work) (Asbestos) (Amendment) Regulations 2013 and MGN 493 (M+F) Exposure to asbestos should be avoided wherever possible, and where it cannot be avoided precautions

Lists of agents, processes and work	Risk	How to avoid risk
	agents) - Asbestosis (scarring of the lung tissue) There are no cures for asbestos related diseases. These diseases can take many years to appear after the period of exposure (long latency). This latency effect means that an exposure occurring at a young age may be a higher risk than the same exposure later in life, simply because a young person is more likely to survive until the time when the disease is most likely to emerge.	should be taken to reduce it to as low a level as possible. Advice is contained in MGN 429 (M+F). Workers should be given any relevant information affecting their health and safety, instruction and training in the safe handling and use of the substance and be provided with adequate supervision within a safe system of work.

Work involving harmful exposure to radiation

Lists of agents, processes and work	Risk	How to avoid risk
Ionising radiation	The risk of developing cancer and hereditary defects from exposure to ionising radiation, which increases slightly for young people, is controlled by setting statutory annual dose limits. According to the HSE website, the main dose limits which relate to the whole body dose are	Regard should be had to the provisions of MGN 197(M+F) Amendment 1 and MGN 451(M+F) In general fishermen may not be exposed to ionising radiation at sea although it is possible that such exposure might

Lists of agents, processes and work	Risk	How to avoid risk
	the most important elements in relation to cancer risk. The limits for young people per calendar year are: - 6 millisieverts (mSv) for trainees under 18 years (30% of the adult limit). Trainees (including students) are defined as being aged 16 years or above receiving instruction or training involving work with ionising radiation. - 1 mSv for employees below 18 years who are not trainees (the same limit as for the general public).	occur when X-ray equipment is being used.
Non-ionising radiation electromagnetic radiation	**Optical radiation:** There is no evidence that young people face greater risk of skin and eye damage than other workers. **Electromagnetic fields and waves:** Exposure within current recommendations is not known to cause ill health to workers of any age. Extreme overexposure to radio-frequency radiation could cause harm by raising body temperature.	Reference documents: Merchant Shipping and Fishing Vessels (Health and Safety at Work) (Artificial Optical Radiation) Regulations 2010 and MGN 428(M+F) Amendment 1 Non-binding EU guidance on artificial optical radiation at work. Seafarers working in hot climates are advised to reduce their exposure to the sun. If it is necessary to work in direct sunlight, appropriate clothing should be worn to protect both head and body. Reference document: Merchant Shipping and

Lists of agents, processes and work	Risk	How to avoid risk
		Fishing Vessels (Health and Safety at Work) (Electromagnetic Fields) Regulations 2016 and MGN 559(M+F). Exposure to electric and magnetic fields should not exceed the restrictions on human exposure published by the Radiation Protection Division of the Health Protection Agency.

Work involving the risk of accidents which it may be assumed cannot be recognised or avoided by young persons owing to their insufficient attention to safety or lack of experience or training

Lists of agents, processes and work	Risk	How to avoid risk
Manufacture and handling of devices, fireworks or other objects containing explosives		It may be necessary to explain the safe handling of pyrotechnics such as distress flares, rockets, line throwing devices, life raft signals and man overboard markers.
Work with fierce or poisonous animals	Unlikely to be relevant to work on ships.	
Animal slaughtering on	Use of knives for filleting and gutting on	

Lists of agents, processes and work	Risk	How to avoid risk
an industrial scale	moving deck etc	
Work involving the handling of equipment for the production, storage or application of compressed, liquefied or dissolved gases.		
Work involving the operation of high risk lifting equipment or acting as signallers to operators of such equipment.	ILO Convention 152 Article 38 There may be substantial risks associated with the use of lifting accessories, for example during ‘slinging’ and employers need to assess whether such work is appropriate for young people.	Reference documents: Merchant Shipping and Fishing Vessels (Provision and Use of Work Equipment) Regulations 2006; Merchant Shipping and Fishing Vessels (Lifting Operations and Lifting Equipment) Regulations 2006 and the respective MGN 331 (M+F) Amendment 1 and MGN 332 (M+F) Amendment 1; Young people may use high-risk lifting machinery under training as long as they are adequately supervised. They should also be supervised after training if considered not sufficiently mature.

Lists of agents, processes and work	Risk	How to avoid risk
Working involving the operation of power machinery or tools	Young people (under 18 years) should not be allowed to use power machinery or tools unless they have the necessary maturity and competence which includes having completed appropriate training.	Young people may operate power equipment and tools during training providing they are sufficiently mature and are adequately supervised. They should also be properly supervised after training until they reach the appropriate level of competence and can work safely unsupervised.
Entry into fish holds. boilers, tanks and cofferdams and other enclosed spaces	Risk from depleted oxygen levels, toxic gases, risk of explosion.	Reference documents: MGN 309(F) Fishing Vessels: the Dangers of Enclosed Spaces Young persons may do this work under training and with adequate supervision.
Handling mooring or tow lines or anchoring equipment	Risk being struck by or striking another person when lines are thrown; risk from parting of ropes and cables under tension.	Reference documents: MGN 592(M+F) which deals with Mooring, Towing or Hauling Equipment On All Vessels – Safe Installation and Safe Operation.
Flammable liquids	Accidental spills can cause fires or explosions Flammable liquids should be used only for their intended purposes: using them for other purposes may lead to fires or explosions	It may be necessary to explain the basics of flammability and what to do if liquid is spilt. It may also be necessary to point out the dangers of using liquids, such as petrol for cleaning machinery.
Flammable gases	Leaking gas from pipes, appliances or	It may be necessary to explain basics of flammability; people need to

Lists of agents, processes and work	Risk	How to avoid risk
	cylinders can cause fires or explosions.	know how to detect leaking gas and what to do in the event of a gas leak. Reference document: MGN 310 (F) FV Risk of Fire and Explosion from Gas Welding and Burning.
Gas cylinders	There is no evidence that young people face greater physical risks from a release of stored energy than other workers. Leaking gas from cylinders may cause fires or explosions. Physical damage to cylinders may cause leaks. Heavy cylinders may cause physical injury if not properly handled. Application of heat to gas cylinders may cause them to burst possibly resulting in “shrapnel” type explosion. Alternatively the contents may be vented through a pressure release valve resulting in fire or explosion.	It may be necessary to explain the basics of flammability; people need to know how to detect leaking gas and what to do in the event of a gas leak. Gas cylinders need to be properly handled, both to avoid the danger of fire or explosion, and the risk of physical injury to the worker, e.g. crushed toes. Gas cylinders need to be safely stored and used away from direct sources of heat. Reference documents: MGN 310 (F) FV Risk of Fire and Explosion from Gas Welding and Burning. Chapter 24, Section 9 and 10 of the Code of Safe Working Practices for Merchant Seafarers.
Work with tanks etc containing chemical agents	Unlikely to apply on fishing vessels.	

Lists of agents, processes and work	Risk	How to avoid risk
Work on or near refrigeration plant	Refrigeration plants on freezer trawlers. Refrigeration gases may be toxic and flammable.	Reference document: Appendix 11 to the Code of Safety for Fishermen and Fishing Vessels 2005 (FAO, ILO and IMO). The risks should be explained. Only to work following instruction and under supervision by a competent person and with safety measures in place, including indicator alarms, with control panel outside the space. In case of any risk of exposure to refrigerant liquid, goggles, rubber or pvc gloves, and other personal protective equipment to be worn.
Handling or taking charge of ship’s boats	Unlikely to apply on fishing vessels	

Working in exposed positions

Lists of agents, processes and work	Risk	How to avoid risk
Working at height, rigging	Risk of falling or dropping items on workers below through lack of concentration or lack of experience	Only to work under supervision. MGN 410 (M+F) Amendment 1 gives general guidance on working at height.
Working at height or over the side.	Risk from falling from aloft or over the ship's side or into the water.	Only to work under supervision. MGN 410 (M+F) Amendment 1 gives general guidance on working at height.
Working on deck in heavy weather.	Risk of injury or of getting washed over the ship's side	Personal Flotation Devices or safety harness (fall prevention) to be worn Reference documents: MGN 588 Amendment 1 Compulsory provision and wearing of personal flotation devices on fishing vessels MSN 1870(M+F) Amendment 3 on the Merchant Shipping and Fishing Vessels (Personal Protective Equipment) Regulations 1999
Work involving risk of structural collapse	May be relevant for cargo stacking and movement of ships' stores	Cargo should be stowed and secured in accordance with the Cargo Securing Manual and where work is to be carried out near a tall stack of cargo or stores, the stack should be secured to prevent it falling. Young workers should only do such work if properly trained or if they are under supervision of a trained person.

Lists of agents, processes and work	Risk	How to avoid risk
Working close to moving machinery, fishing gear and mooring operations	Risk from fishing gear and mooring lines.	Ensure awareness of where to stand/work on the fishing vessel where fishermen to avoid risk of entrapment, being struck by gear etc (i.e. safe zones) including avoiding snap back zones from mooring gear.

Operating hazardous equipment

Lists of agents, processes and work	Risk	How to avoid risk
Servicing of electrical equipment	Risk of electric shock or other injury if equipment not properly isolated from a power source.	Reference documents: Merchant Shipping and Fishing Vessels (Provision and Use of Equipment) Regulations 2006 and MGN 331(M+F) Amendment 1
Work involving high-voltage electrical hazards.	The risk is one of electric shock, burns or electrocution. There is no evidence that young people face greater physical risk from electricity than other workers.	Reference documents: MGN 452(M). As with adults, young people should not undertake work involving electricity unless they have the necessary knowledge and/or experience to prevent danger or injury; or are under an appropriate level of supervision having regard to the nature of the work.

Lists of agents, processes and work	Risk	How to avoid risk
Cleaning of catering machinery	Risk of injury from moving or sharp parts	Regard should be had to the provisions of the Merchant Shipping and Fishing Vessels (Provision and Use of Equipment) Regulations 2006 and MGN 331 (M+F) Amendment 1. Unless trained and properly supervised young persons must not clean a machine with dangerous parts.
Use of ship’s laundry equipment	Industrial laundry equipment unlikely to apply on fishing vessels. See above on risk from electrical equipment – noting additional risk of damp environment.	

Work in which there is a risk to health from extreme cold or heat

Lists of agents, processes and work	Risk	How to avoid risk
Extremes of cold or heat	Exposure to extreme cold carries risks to workers of all ages. These are principally hypothermia and local cold injury (frostnip/frostbite). People of all ages vary in their ability to tolerate cold conditions. Exposure to extreme heat carries risks for workers of all ages. These	Depending on the findings of the risk assessment, the provision of appropriate protective clothing and control of periods of exposure will help to minimise

Lists of agents, processes and work	Risk	How to avoid risk
	include collapse due to heat exhaustion or potentially fatal heat stroke. Protective clothing may exacerbate the problem by preventing the body from losing heat normally. There are no special considerations for young people – their response to work in hot conditions will depend on physical fitness, physique and past experience of hot conditions, which will be variable.	the risk. Any intended exposure to heat must be carefully assessed and the risks can be minimised by measures such as introducing suitable work patterns, prior medical assessment of workers and proper supervision of the work.